Lecture 1: Introduction to GNBT 120

Dan Koenig

Instructor

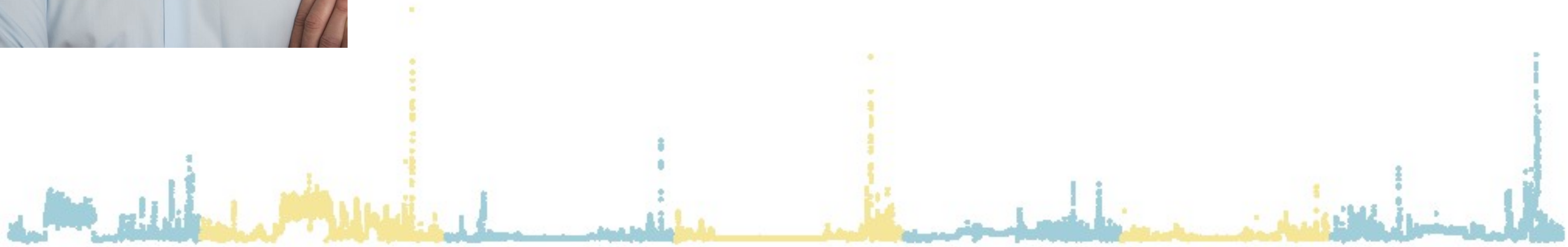
Dan Koenig

dkoenig@ucr.edu

Assistant Professor of Genetics

Department of Botany and Plant Sciences

Institute of Integrative Genome Biology



A new course, and new ideas

This is the second offering of this course, and there are bound to be some bumps in the road, I am committed to making this a great experience for you

This course may cover ideas and methods that are very new to you

It will take some time to gain confidence with these new methods and ideas, but you will emerge from the course with a foundation in the field and skills that are not only useful to biologists

Attribution

Much of the beginning of this course is modelled and directly uses materials from Julin Maloof's BIS180 course at UC Davis. We will use some of the labs from that course to begin to develop your skills.

Course Objectives

To expose you to common methods and thinking used in Computational Biology.

After completing this course you will be able to:

- Connect to a high performance compute cluster
- Use basic command line skills to navigate directories, create and manipulate files, and to execute programs
- Iterate processes using for loops
- Keep a computational laboratory notebook using Markdown
- Manage a reproducible computational project using GitHub
- Conduct analysis of biological datasets
- visualize results using the statistical computing software R and the RStudio interface

Course Objectives

In addition to developing basic computational skills, you will conduct biological research that may encompass some of the following:

- Assemble a new genome
- Identify the sequences within the genome with particular biological function
- Align sequences to a genome and identify genetic differences between individuals
- Align entire genomes to one another and identify large scale changes
- Link genomic differences to changes in phenotype using genome-wide association studies
- Identify changes in gene expression between samples using transcriptomic datasets

How to succeed in this course?

1. Take your time to understand what you are doing
2. Make sure you read the instructions carefully
3. It is better to type out commands rather than copy and paste them
4. Ask questions when you need help!
5. Help each other

Course organization

Lecture/Lab:

Tuesdays and Thursdays, 12-4

I will deliver the lectures at the beginning of the period

We will use the rest to work on assignments and lab protocols

Evaluation

- Lab exercises 40%
    a. Assignments will be turned in via github
    b. There will be a separate repo for each assignment
    c. Detailed instructions are given in the lab page for today
    d. I can help if you get stuck
    e. For today's lab you will need to turn in four files (two markdown files and their corresponding .html counterparts).

Evaluation

- Lab exercises 40%
- Quizzes 10%
    a. Short quizzes will be administered weekly. These will be 10-15 mins multiple choice quizzes focused on major concepts from the lecture/lab/reading.

Evaluation

- Lab exercises 40%
- Quizzes 10%
- Participation 10%
    a. Attendance is critical to your success in this course. If you are sick, let me know and don't come, but otherwise you will receive points for every time you make it to class.

Evaluation

- Lab exercises 40%
- Quizzes 10%
- Participation 10%
- Lab Project 40%
    a. You will conduct a real research project in this course using a dataset produced in my lab
    b. You will use the methods that you learn to analyze a genomic dataset, you will present your findings in the last meeting of the course, and you will turn in a lab report on your findings and report all of the code necessary to reproduce your analyses.
    c. This project takes the place of exams (there are no exams) so it is important that you make sure to put together a good product
    d. More information on the project will be provided soon.

Course Website

Additional information can be found on the course website:

https://dpkoenig.github.io/GNBT120/

1. Syllabus
2. Lab materials (will be updated)
3. Course Schedule (will be updated)
4. Assignments (will be updated)

The canvas site will be used for announcements, quizzes and grade reporting

Quick tour and course schedule discussion

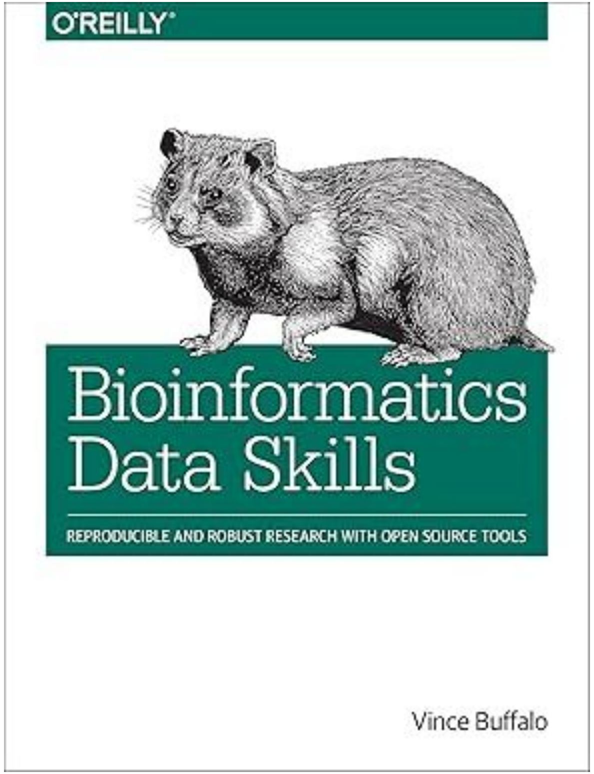
Do your own work

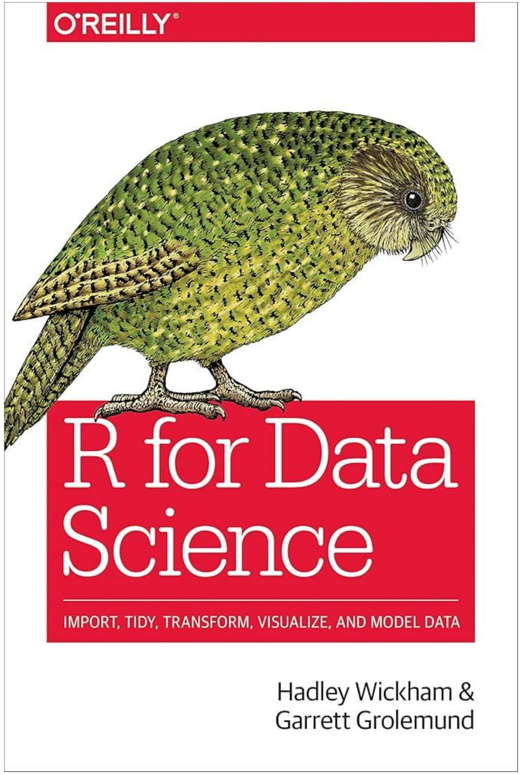
Developing code is an interactive process. Both your friends and the web can be excellent resources.

However Any direct copying of text or code **from any person (in this class or on the web, etc)** is considered plagiarism in the context of this course.

If you receive inspiration or ideas from an external source **give attribution**

Texts


O'REILLY
Bioinformatics
Data Skills
REPRODUCIBLE AND ROBUST RESEARCH WITH OPEN SOURCE TOOLS
Vince Buffalo


O'REILLY
R for Data
Science
IMPORT, TIDY, TRANSFORM, VISUALIZE, AND MODEL DATA
Hadley Wickham &
Garrett Grolemund

The goals of genomics

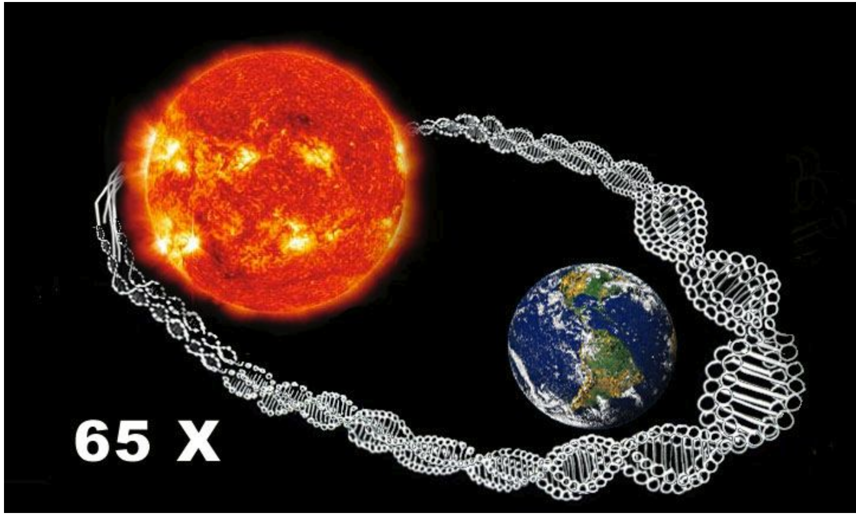
How does this simple code specify a complex organism?

How do changes in the code translate into biological diversity?

Photo: American Museum of Natural History
https://www.flickr.com/photos/mukluk/440494350/ CC-SA

There is a lot of it!

- The human genome contains 3.2 billion bases of DNA and there are 2 copies per cell
- A single nucleotide is ~ 340 picometers
- So, the length of all the DNA in a single cell is a about 2 meters long if stretched out
- There are somewhere around 37.2 trillion cells in your body
- This means that the total length of DNA in your body is ~ $2*10^{13}$ m long, and could reach to the sun and back about 65 times!



We live in an amazing time

536 R. WU AND A. D. KAISER

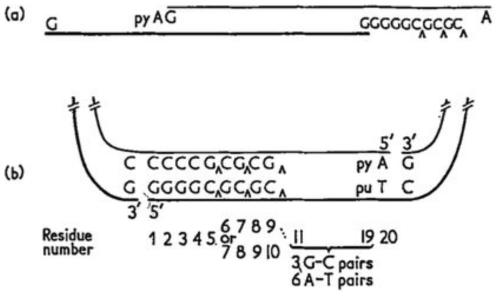


FIG. 7. Base sequence of the cohesive ends.

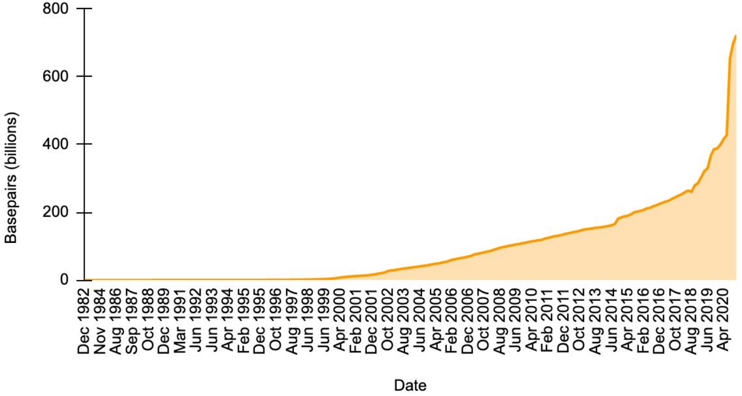
(a) The structure of a native, linear λ DNA molecule is represented by 2 lines, differentiated by thickness as in Fig. 5. The thick line designates the *l*-strand and the thin line the *r*-strand. The 3′- and 5′-terminal nucleotides are identified. The nucleotides, added in the presence of DNA polymerase, which have been identified, are written beyond the ends of the lines. An additional dG residue is located at one of the three caret marks.

(b) That portion of a circular molecule which contains the cohered ends is shown. The base sequence is based on (a), the complementarity of the two protruding single strands, and on the total composition of the cohesive ends. An additional dG residue is located at one of the caret marks above the thicker line and an additional dC residue at the corresponding caret below the thinner line.

The residues are numbered starting with the 5′-terminus of the left end.

50 years ago

Sequence in national repositories



We live in an amazing time

536 R. WU AND A. D. KAISER

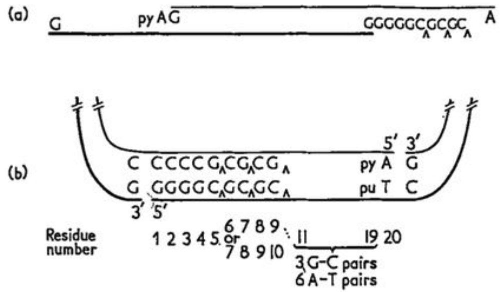


Fig. 7. Base sequence of the cohesive ends.

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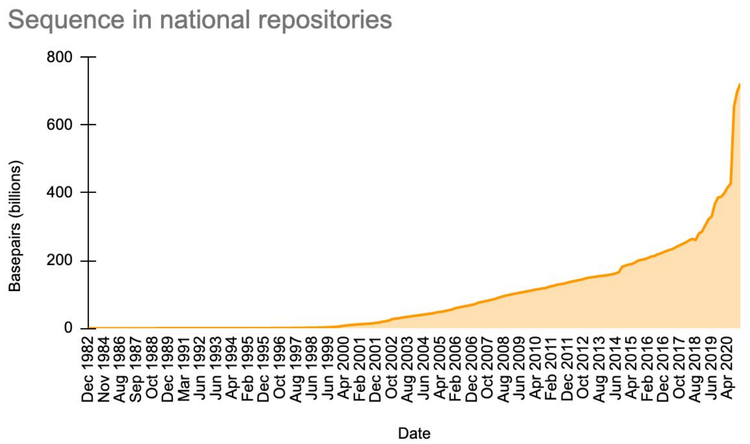
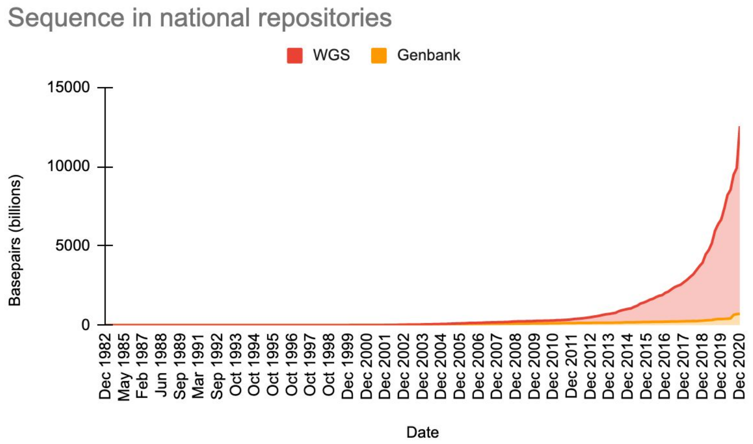




What can we do with this data?

Bioinformatics is the branch of biology that deals with in silico, meaning “in the computer,” processing and analysis of DNA, RNA and protein sequence data.

Bio = Biology

Informatics = The process of data storage and retrieval

Texas Advanced Supercomputing Center

How do we store and use the data?

Sequence data are stored in public databases around the world

The three major sites are the DNA databank of Japan (NIG), EMBL (European Bioinformatics Institute, and GenBank (NCBI)

You will make use of DNA databases in the lab section

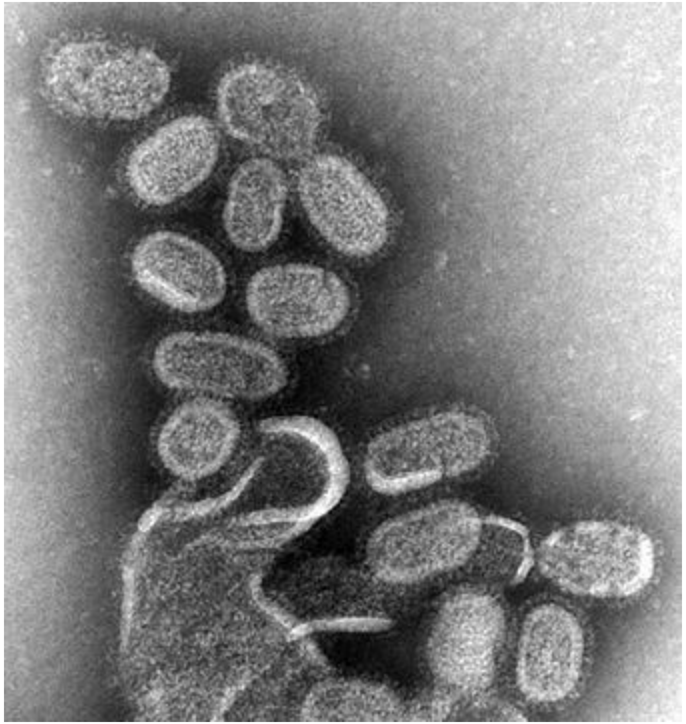
You can be a genomicist!

These data are freely available to the public and can be analyzed by anyone with a computer and internet connection

The flu of 1918

Killed an estimated 50,000,000 people (3% of the world’s population), making it one of the worst natural disasters in history

Influenzavirus

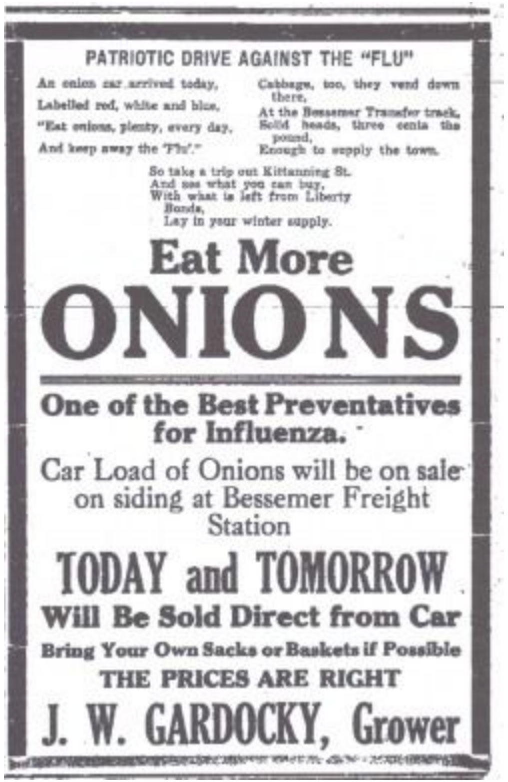


- RNA virus
- Three human relevant forms A, B and C
- A strains are the most virulent sources of human infection
- A strains include H1N1, the strain responsible for the spanish flu
- Responsible for 3-5 million severe illnesses and hundreds of thousands of deaths every year

How to stop flu pandemics?

Flu Vaccination programs

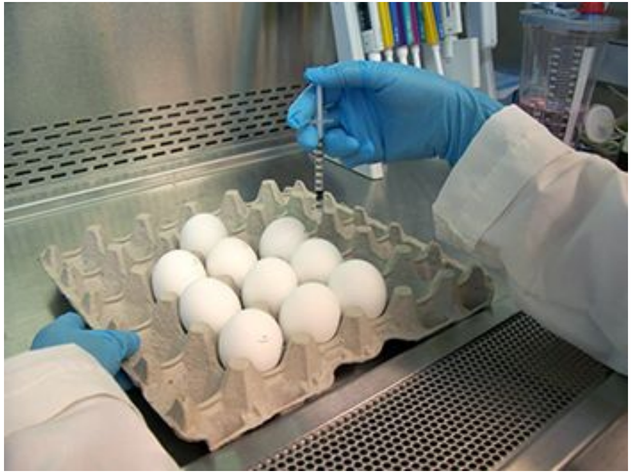
- The flu rapidly mutates allowing the same strain to reinfect the same individuals in subsequent years
- This is a challenge for vaccination programs: new vaccines must be generated each year against several strains
- The WHO attempts to predict the predominant strains and makes recommendations each year prior to the flu season



How to stop flu pandemics?

Flu Vaccination programs

- Predicting the correct strains is very difficult
- Laboratory assays of Flu vaccine effectiveness are important, but they do not predict of Flu strain success
- Flu vaccine effectiveness ranges from 10-56% depending on the year according to the CDC https://www.cdc.gov/flu/vaccines-work/effectiveness-studies.htm



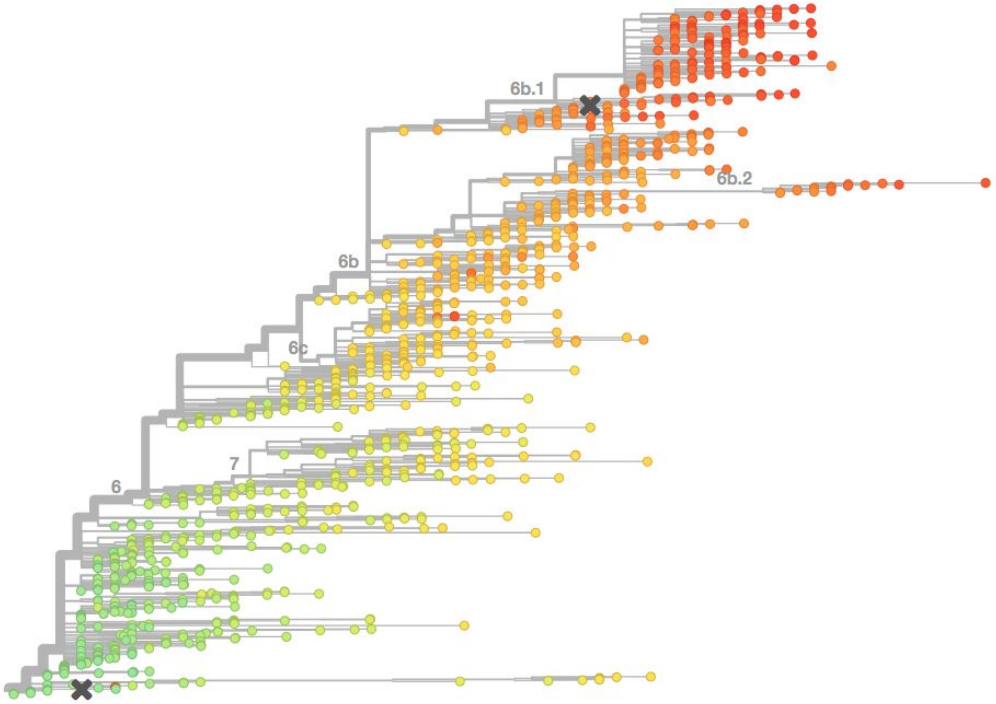
An emerging Solution

Researchers are combining the power of cheap, fast DNA sequencing with evolutionary analysis to generate real-time prediction of flu strain spread

These approaches show early promise in complementing previous strategies, and can be done in real-time

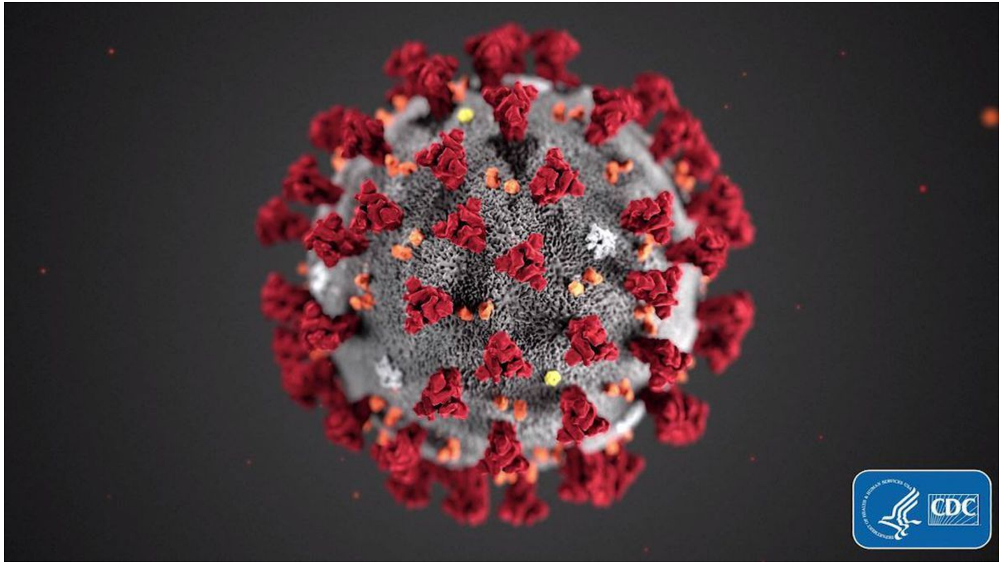
nextstrain

Real-time tracking of virus evolution



H1N1 Flu: http://nextflu.org/h1n1pdm/12y/

The rise and spread of COVID-19



- Single stranded RNA genome encased in protein
- The genome is about 30,000 nucleotides long (less than 1/100000th of the human genome)
- Similar to some types of the common cold, but also to other deadly diseases

The rise and spread of COVID-19

[Nextstrain / ncov / gisaid / global]

Plan for Today

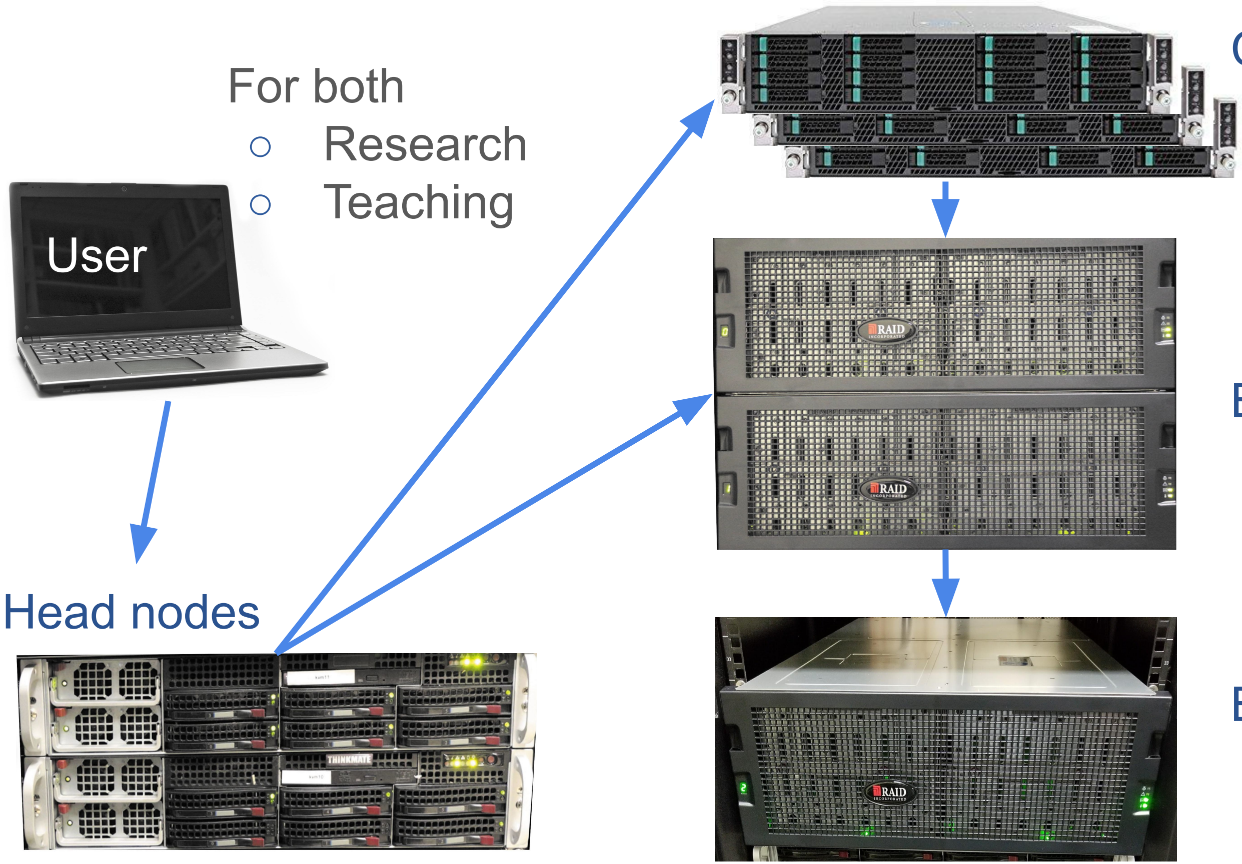
1. Connect to the compute cluster we will use throughout the course
2. Setup Git Authentication
3. Learn how to use Git to turn in projects/assignments
4. Introduce Markdown

What Is a Computer Cluster?

- A computer cluster is an assembly of CPU units, so-called computer nodes that work together to perform many computations in parallel.
- An internal network connects the nodes to a larger unit, while a head node controls the load and traffic across the entire system.
- Usually, users log into the head node to submit their computer requests via `srun` or `squeue` to a queuing system provided by resource management and scheduling software, such as Slurm. The queuing system distributes the processes to the computer nodes in a controlled fashion.

Architecture of the HPC Cluster (from Thomas Girke)

For both
- Research
- Teaching



Computer cluster

- 106 Nodes (Intel, AMD, GPU)
- ~8500 CPU cores
- 512-1024GB (4TB) RAM per node
- GPU: 6x K80, 2x P100, 24x A100
- IB network @ 56-400Gbs

Big data storage cluster

- ~4PB GPFS storage (scales to >50PB)
- Home directories on dedicated system

Backup system

- ~4PB GPFS storage
- Separate server room

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What is Git?

- Git is a system for storing software and for version control
- It allows you to keep track of all changes that have occurred during your work and to return to previous versions if necessary
- It is essential to transparent and reproducible computational research across many disciplines
- You will turn in your assignments via GitHub and we will set this up today

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What is Markdown and why do we use it?

Markdown is a text-based formatting system for quickly and easily generating nicely formatted output.

It helps achieve three guiding principles:

1. Clear documentation
2. Reproducibility
3. Open formats

Markdown vs docx

What is we want to produce this:

A markdown example

bulleted list

- point 1
- point 2
- point 3

some **bolded** or *italicized* text

code chunk

```
#simple code
x <- 1:10
y <- 1:10
x + y
```

The markdown file that generates it is

```
# A markdown example

## bulleted list

* point 1
* point 2
* point 3

some __bolded__ or _italicized_ text

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